

Foundations of Computer Vision

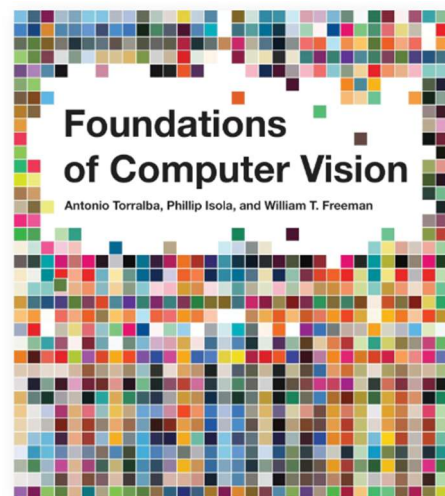
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Agenda

- Course content
 - What is computer vision?
- Some course logistics



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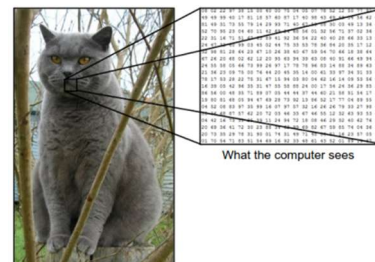
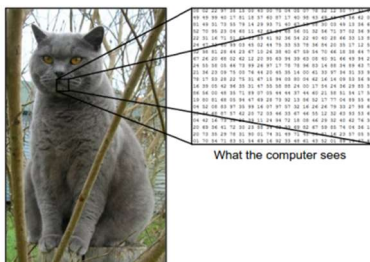
How to define Computer Vision?

- Central goal in computer vision: Endow eyes to a computer.

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Defining Computer Vision?

- Endowing **1. Recognition**: Attaching semantic category labels
 - Later, semantic labels to activities

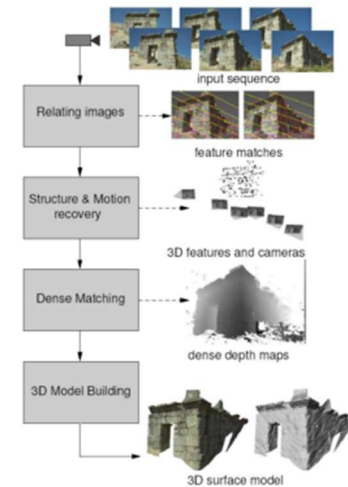


Looks like a cat

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Defining Computer Vision?

- Endowing **2.Recovery**
- 3D geometry of the world
 - Correspondence (match images to images, images to 3D)
 - Reconstruction (recover the structure)
 - Create novel views (reproject)



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Defining Computer Vision?

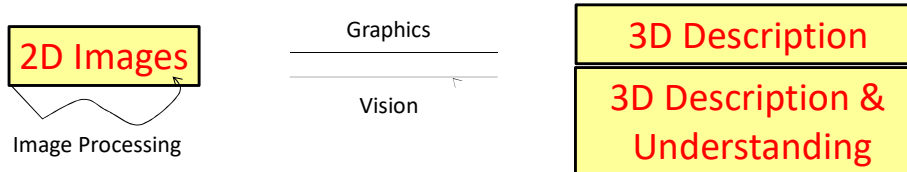
- Endowing: **3. Reorganization**
- Group the visual elements so as to determine the meaning as a whole



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Scope of Computer Vision

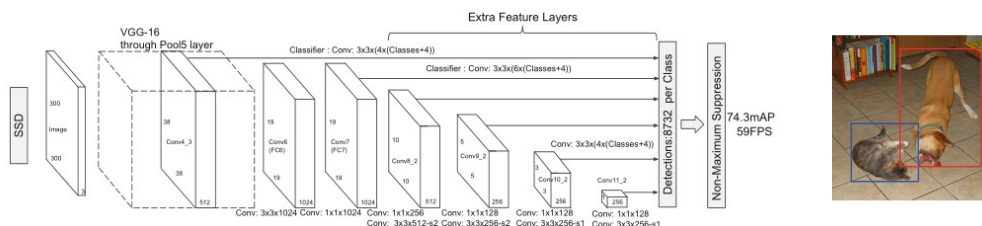
- Relationship to other courses:
 - image processing (fundamentals, advanced, medical),
 - graphics,
 - pattern recognition,
 - photogrammetry,
 - robotics



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Scope of Computer Vision

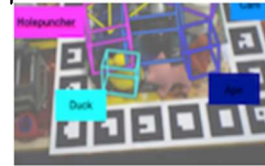
- Many recognition tasks are handled by pattern matching and arguably are solved problems to some extent
 - Deep-learning is the messiah (or culprit)
 - That's not to say that they are not important, but it's unclear if the content is "deep-learning" or "core computer vision"



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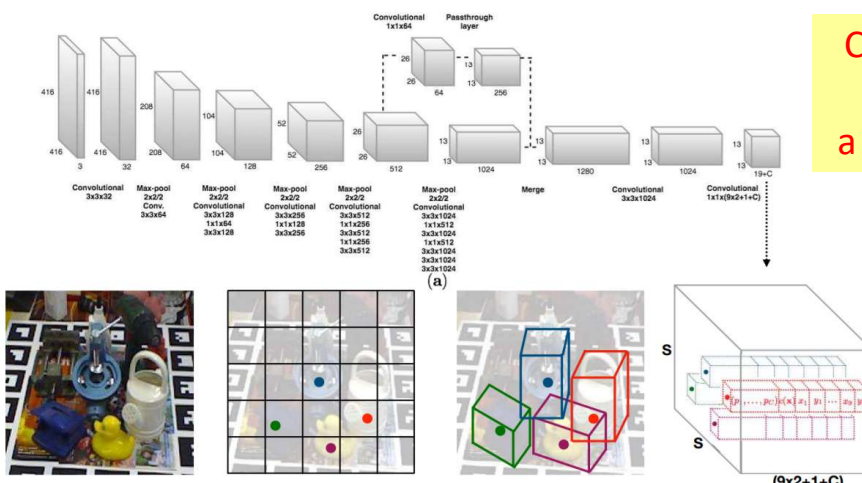
Scope

- Goal: Introduce a set of basic computational techniques aimed at estimating or making explicit the geometric properties with one or more cameras



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Realtime Single Shot 6D Pose



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Scan



We are unable to read the text in the white region clearly?

Much better!

Concept:
Homography

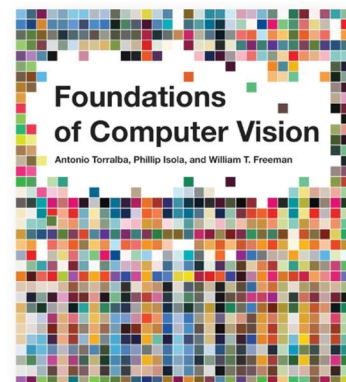


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Summary

- Course content
 - What is computer vision?
- Some course logistics
 - Please attend all classes
 - Please keep the camera on
 - Exams are key to passing the course, but
 - attending all classes is the way to get there
 - Assignments

Please try to attend all classes



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